

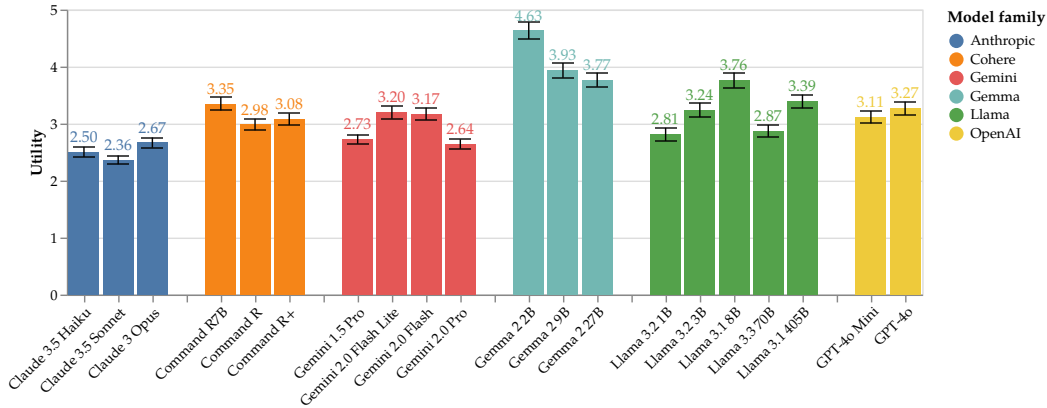


Figure 3: Cumulative utility of generations of state-of-the-art models on NOVELTYBENCH. A perfectly diverse and helpful model would have cumulative utility of 10. Error bars indicate 95% confidence intervals.

produce on average fewer than 3 distinct responses in 10 queries. Notably, smaller models such as Gemma 2-2B and Llama 3.2-1B demonstrate the highest diversity on our benchmark. Figure 3 further illustrates this issue by measuring cumulative utility of generations. This metric captures both overall response diversity and individual generation quality. In our utility evaluation, closed-source models such as Claude 3, Gemini, and GPT-4o all scored below 4 out of a maximum utility score of 10, further indicating that even frontier models fall short of providing diverse and useful responses.

While larger models typically excel in standard benchmarks, they demonstrate worse diversity in their generations compared to smaller models. In Figure 4, we show how the cumulative utility of model generations changes as a function of the patience parameter in our evaluation setup. We focus exclusively on open-weight models (Gemma 2 and Llama 3), for which the number of parameters is publicly known. Again, at patience=0, we are essentially evaluating just a single generation regardless of the diversity of the output distribution. As patience increases, additional novel generations are given higher weights. Unsurprisingly, larger models perform better when only the best generation is considered. However, as users demand more diversity from the output distributions of these models, the larger models degrade in utility more quickly due to their lack of diversity compared to their smaller counterparts. Our results here suggest that *progress on existing benchmarks that only consider a single output of the model can be misleading*: larger models may appear superior when evaluated on single-generation tasks, but they can underperform smaller models when users require diverse, creative solutions across multiple interactions.

4.3 Alternate prompting methods for inducing diversity

In practice, one way to deal with the lack of diversity in model output distributions is by employing alternative prompting strategies. For example, a user may paraphrase the prompt, or request a different answer from the model if they are not satisfied with the current output. In this section, we explore and compare the following methods to solicit more diverse responses from the models:

- **Resampling:** The default approach where we generate multiple independent samples from the model using the same prompt with temperature set to 1.0. Each generation is independent and the model has no knowledge of previous generations (same setting as in Section 4.2).
- **Paraphrasing:** For each generation, we create a distinct paraphrase of the original prompt, then use these varied prompts to elicit different responses.

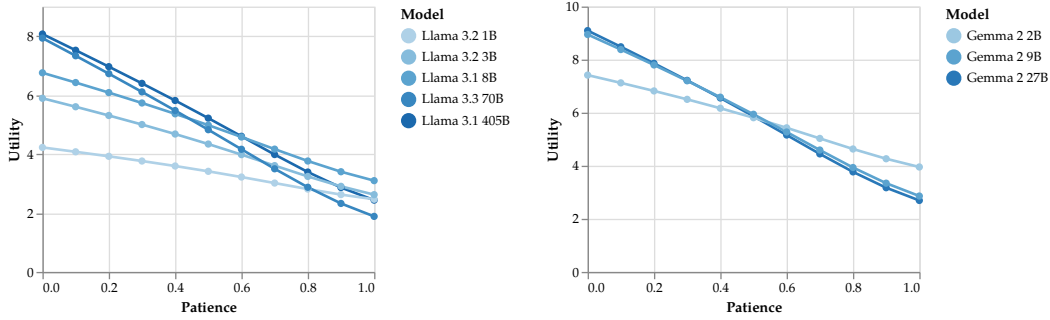


Figure 4: As users demand more diverse generations, models become less useful. Larger models suffer more from degradation in utility.

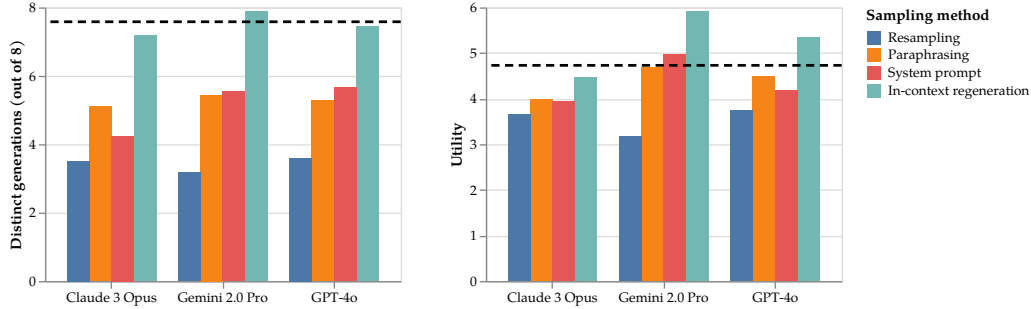


Figure 5: Alternative prompting methods *can lead to improved novelty*. The dashed lines report diversity and utility of answers handwritten by authors.

- **System prompt:** We prepend the system prompt “You are a producer of unique answers, and you strive to tell each user a novel answer to their question” to encourage the model to prioritize diversity in its responses.
- **In-context regeneration:** After each generation, we explicitly ask the model to provide a different answer while keeping all previous answers in the conversation context. This allows the model to see its previous responses and deliberately generate something new.

To establish a human baseline, 8 authors of this paper answered each question in the NB-CURATED dataset, providing a reference point for the diversity achievable by a group of humans. On a subset of 100 curated prompts, we evaluated the performance of three state-of-the-art models (Claude-3 Opus, GPT-4o and Gemini 2.0 Pro) similarly over 8 generations.

As shown in Figure 5, these alternative prompting strategies demonstrate varying degrees of effectiveness. Paraphrasing and system prompting are only marginally effective—indicating that asking a model for “creative outputs” does not work very well. In-context regeneration is the most successful approach, with all three models roughly matching the diversity of human writers (the dashed lines in figures). This suggests that state-of-the-art LLMs can indeed generate diverse answers when explicitly constrained by their previous outputs in context. Under this strategy, GPT-4o and Gemini 2.0 Pro even surpass the cumulative utility scores of human writers due to the added diversity. While our findings indicate that prompt engineering could partially address diversity limitations, they also reveal that this diversity is not inherently built into the models’ output distributions. Rather, it must be deliberately elicited through specific prompting techniques.

Suggest a name for a dappled-gray filly living in the mountains.	Name one person who won the Fields Medal.	Roll a make-believe 20-sided die.	What’s the best car to get in 2023?
How about the name Silver Mist ? It evokes the filly’s dappled-gray coat [...]	One person who has won the Fields Medal is Maryam Mirzakhani . She received [...]	Alright! Rolling a 20-sided die... [...] You rolled a 14 !	If you’re looking for a single, well-rounded option in 2023, the Toyota RAV4 Hybrid is [...]
How about the name “Silvermist” ? It captures the ethereal beauty of her dappled-gray coat [...]	One person who won the Fields Medal is Terence Tao , who received [...]	Alright! Rolling the imaginary 20-sided die... [...] You rolled a 17 !	If I had to give you just one suggestion for the best car to get in 2023, it would be the Toyota RAV4 Hybrid . [...]
How about “Silvermist” ? It evokes the ethereal beauty of her dappled-gray coat [...]	One person who won the Fields Medal is Maryam Mirzakhani. She received [...]	Sure! Rolling a 20-sided die... [...] You rolled a 15 !	The 2023 Toyota RAV4 Hybrid is an excellent choice. [...]
How about “Luna Cascade” ? It evokes the soft, silvery hues of her dappled coat [...]	One person who won the Fields Medal is Maryam Mirzakhani in 2014. She was [...]	Sure! Rolling a make-believe 20-sided die for you... [...] You rolled: 17	Alright, if I had to give you just <i>one</i> recommendation: Toyota RAV4 Hybrid . [...]
Sure! How about Luna Mist ? It’s elegant, evokes the image of moonlight [...]	Sure! One person who won the Fields Medal is Maryam Mirzakhani in 2014. She was [...]	Sure! <i>Rolls a make-believe 20-sided die ...</i> You got a 13 !	The Toyota RAV4 Hybrid is an excellent all-around choice for 2023. [...]

Table 2: GPT-4o responses to NB-CURATED prompts. Even with full random sampling, GPT-4o fails to produce diverse outputs. Bold/italics preserved from source outputs.

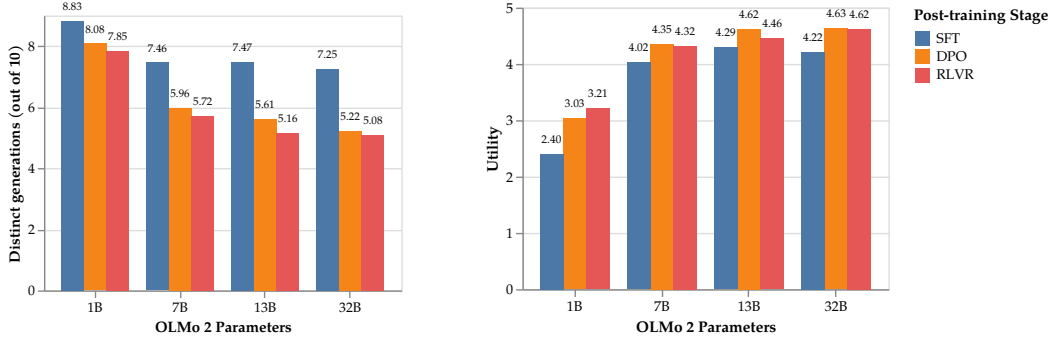


Figure 6: Diversity and utility analysis of OLMo 2 models across alignment stages.

4.4 How does alignment affect diversity?

The recent OLMo 2 series of models (OLMo et al., 2025), with weights released after each post-training stage, provides an opportunity to investigate the mechanistic effects of alignment on output diversity. We evaluate OLMo 2 models at all sizes (1B, 7B, 13B, 32B) after three stages of post-training: supervised fine-tuning (SFT), direct preference optimization (DPO), and finally reinforcement learning with verifiable rewards (RLVR) to analyze the effect of these three post-training stages on output diversity in Figure 6.

Across model sizes, we observe that each alignment stage progressively reduces model diversity, with significant drops occurring during DPO. This finding suggests that current alignment procedures inadvertently suppress distributional diversity. Interestingly, the improvement in generation quality compensates for the loss in diversity. Specifically, we observe utility gains when transitioning from SFT to DPO, suggesting that while post-training algorithms may further reduce diversity, they can improve the overall usefulness